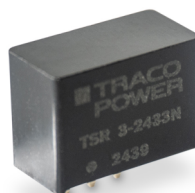


## Non-Isolated DC/DC Converter (POL)

## TSR 3N Series, 3 A

- Highly cost-efficient design
- Compact SIP-3 package:  
14 x 7.6 x 10.2 mm
- Pin compatible with TO-220 package  
linear regulators
- Excellent thermal capabilities
- Operation temperature. range  $-40^{\circ}\text{C}$  to  $+90^{\circ}\text{C}$
- Efficiency up to 95%
- Wide input operating range 4.6-36 VDC
- Short circuit protection, current  
limitation and under voltage lockout  
features
- Excellent line / load regulation
- 3-year product warranty



The TSR 3N is a 3 Amp step-down switching regulator series and a drop-in replacement for any TO-220 package linear regulators. It comes in a compact SIP-3 plastic package and complements our new generation of POL converters focusing strongly on a cost-efficient design while also improving on critical electrical specifications. The Input ranges from 4.6 to 36 VDC and output voltages are available between 1.2 and 15VDC. Depending on the model, the efficient design allows full load operation up to  $+90^{\circ}\text{C}$  ambient temperature (at nominal  $V_{in}$ ) without the need of any heat sink or forced cooling. The TSR 3N series provides other significant features like short circuit protection, current limitation and under voltage lockout. Overall, it offers a broad application range in many environments and is especially suited for high volume projects where the series will help to reduce production cost by delivering not only a highly cost efficient but also reliable solution.

| Models       |                        |                            |                        |                    |
|--------------|------------------------|----------------------------|------------------------|--------------------|
| Order Code   | Output Current<br>max. | Input Voltage<br>Range     | Output Voltage<br>nom. | Efficiency<br>typ. |
| TSR 3-2412N  | 3'000 mA               | 4.6 - 28 VDC (12 VDC nom.) | 1.2 VDC                | 83 %               |
| TSR 3-2415N  |                        | 4.6 - 32 VDC (12 VDC nom.) | 1.5 VDC                | 85 %               |
| TSR 3-2418N  |                        | 4.6 - 36 VDC (12 VDC nom.) | 1.8 VDC                | 86 %               |
| TSR 3-2425N  |                        |                            | 2.5 VDC                | 89 %               |
| TSR 3-2433N  |                        | 6.5 - 36 VDC (12 VDC nom.) | 3.3 VDC                | 90 %               |
| TSR 3-2450N  |                        |                            | 5 VDC                  | 92 %               |
| TSR 3-2465N  |                        | 9 - 36 VDC (12 VDC nom.)   | 6.5 VDC                | 93 %               |
| TSR 3-2490N  |                        | 11 - 36 VDC (24 VDC nom.)  | 9 VDC                  | 93 %               |
| TSR 3-24120N |                        | 15 - 36 VDC (24 VDC nom.)  | 12 VDC                 | 94 %               |
| TSR 3-24150N |                        | 18 - 36 VDC (24 VDC nom.)  | 15 VDC                 | 94 %               |

## Input Specifications

|                        |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input Current          | - At no load   | 12 Vin models: 1 mA typ.<br>24 Vin models: 1 mA typ.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                        | - At full load | 12 Vin models: 370 mA typ. / 954 mA max. (1.2 Vout model)<br>449 mA typ. / 1'158 mA max. (1.5 Vout model)<br>533 mA typ. / 1'374 mA max. (1.8 Vout model)<br>719 mA typ. / 1'855 mA max. (2.5 Vout model)<br>939 mA typ. / 2'421 mA max. (3.3 Vout model)<br>1'383 mA typ. / 2'527 mA max. (5 Vout model)<br>1'789 mA typ. / 2'539 mA max. (6.5 Vout model)<br>24 Vin models: 1'232 mA typ. / 2'645 mA max. (9 Vout model)<br>1'625 mA typ. / 2'572 mA max. (12 Vout model)<br>2'031 mA typ. / 2'679 mA max. (15 Vout model) |
| Under Voltage Lockout  |                | 12 Vin models: 2.2 VDC min.<br>24 Vin models: 2.2 VDC min.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Recommended Input Fuse |                | 24 Vin models: 7'000 mA (fast acting)<br>1.2 Vout models: 3'000 mA (fast acting)<br>1.5 Vout models: 3'500 mA (fast acting)<br>1.8 Vout models: 4'000 mA (fast acting)<br>2.5 Vout models: 5'000 mA (fast acting)<br>3.3 Vout models: 7'000 mA (fast acting)<br>5 Vout models: 7'000 mA (fast acting)<br>6.5 Vout models: 7'000 mA (fast acting)<br>(The need of an external fuse has to be assessed in the final application.)                                                                                              |
| Input Filter           |                | Internal Capacitor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Output Specifications

|                                        |                                 |                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Voltage Set Accuracy                   |                                 | ±2% max.                                                                                                                                                                                                                                                                                                                                                         |
| Regulation                             | - Input Variation (Vmin - Vmax) | 0.5% max.                                                                                                                                                                                                                                                                                                                                                        |
|                                        | - Load Variation (0 - 100%)     | 2% max.                                                                                                                                                                                                                                                                                                                                                          |
| Ripple and Noise<br>(20 MHz Bandwidth) |                                 | 24 Vin models: 75 mVp-p typ. (w/ 10 µF)<br>1.2 Vout models: 50 mVp-p typ. (w/ 10 µF)<br>1.5 Vout models: 50 mVp-p typ. (w/ 10 µF)<br>1.8 Vout models: 50 mVp-p typ. (w/ 10 µF)<br>2.5 Vout models: 50 mVp-p typ. (w/ 10 µF)<br>3.3 Vout models: 50 mVp-p typ. (w/ 10 µF)<br>5 Vout models: 50 mVp-p typ. (w/ 10 µF)<br>6.5 Vout models: 75 mVp-p typ. (w/ 10 µF) |
| Capacitive Load                        |                                 | 1.2 Vout models: 2'200 µF max.<br>1.5 Vout models: 2'200 µF max.<br>1.8 Vout models: 1'500 µF max.<br>2.5 Vout models: 1'500 µF max.<br>3.3 Vout models: 1'200 µF max.<br>5 Vout models: 820 µF max.<br>6.5 Vout models: 620 µF max.<br>9 Vout models: 620 µF max.<br>12 Vout models: 470 µF max.<br>15 Vout models: 470 µF max.                                 |
| Minimum Load                           |                                 | Not required                                                                                                                                                                                                                                                                                                                                                     |
| Temperature Coefficient                |                                 | ±0.02 %/K max.                                                                                                                                                                                                                                                                                                                                                   |
| Hold-up Time                           |                                 | 20 ms min.                                                                                                                                                                                                                                                                                                                                                       |
| Short Circuit Protection               |                                 | Continuous, Automatic recovery                                                                                                                                                                                                                                                                                                                                   |
| Output Current Limitation              |                                 | 260% typ. of Iout max. (1.2 - 3.3 Vout models)<br>240% typ. of Iout max. (other Vout models)                                                                                                                                                                                                                                                                     |

All specifications valid at nominal voltage, resistive full load and +25°C after warm-up time, unless otherwise stated.

## Safety Specifications

|                       |                             |                                                |
|-----------------------|-----------------------------|------------------------------------------------|
| Standards             | - IT / Multimedia Equipment | Designed for IEC/EN/UL 62368-1 (not certified) |
| Pollution Degree      |                             | PD 2                                           |
| Over Voltage Category |                             | Not mains connected                            |

## EMC Specifications

|                 |                           |                                                                                                |
|-----------------|---------------------------|------------------------------------------------------------------------------------------------|
| EMI (Emissions) | - Conducted Emissions     | EN 55032 class A (with external filter)                                                        |
|                 | - Radiated Emissions      | EN 55032 class B (with external filter)                                                        |
|                 |                           | EN 55032 class A (with external filter)                                                        |
|                 |                           | EN 55032 class B (with external filter)                                                        |
|                 | External filter proposal: | <a href="http://www.tracopower.com/tsr3n-emc-filter/">www.tracopower.com/tsr3n-emc-filter/</a> |
| EMS (Immunity)  | - EFT (Burst) / Surge     | EN 61000-4-4, $\pm 2$ kV, perf. criteria A                                                     |
|                 |                           | EN 61000-4-5, $\pm 2$ kV, perf. criteria A                                                     |
|                 | Ext. input component:     | 1500 $\mu$ F / 100 V                                                                           |

## General Specifications

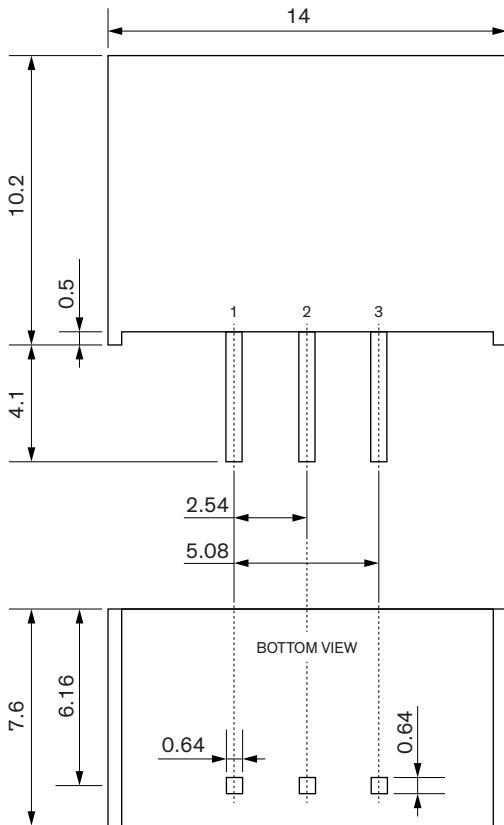
|                          |                         |                                                                                                                                                                           |
|--------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relative Humidity        |                         | 95% max. (non condensing)                                                                                                                                                 |
| Temperature Ranges       | - Operating Temperature | -40°C to +90°C                                                                                                                                                            |
|                          | - Case Temperature      | +110°C max.                                                                                                                                                               |
|                          | - Storage Temperature   | -55°C to +125°C                                                                                                                                                           |
| Power Derating           | - High Temperature      | Depending on model                                                                                                                                                        |
|                          |                         | See application note: <a href="http://www.tracopower.com/tsr3n-cc/">www.tracopower.com/tsr3n-cc/</a>                                                                      |
| Cooling System           |                         | Natural convection (20 LFM)                                                                                                                                               |
| Regulator Topology       |                         | Buck Converter                                                                                                                                                            |
| Switching Frequency      |                         | 380 kHz typ. (PWM) (1.2-2.5 Vout models)                                                                                                                                  |
|                          |                         | 850 kHz typ. (PWM) (3.3-6.5 Vout models)                                                                                                                                  |
|                          |                         | 1000 kHz typ. (PWM) (9-15 Vout models)                                                                                                                                    |
| Insulation System        |                         | Non-isolated                                                                                                                                                              |
| Reliability              | - Calculated MTBF       | 1'600'000 h (MIL-HDBK-217F, ground benign)                                                                                                                                |
| Washing Process          |                         | According to Cleaning Guideline<br><a href="http://www.tracopower.com/info/cleaning.pdf">www.tracopower.com/info/cleaning.pdf</a>                                         |
| Environment              | - Vibration             | MIL-STD-810F                                                                                                                                                              |
|                          |                         | MIL-STD-202                                                                                                                                                               |
| Housing Material         |                         | Plastic (UL 94 V-0 rated)                                                                                                                                                 |
| Base Material            |                         | Non-conductive Plastic (UL 94 V-0 rated)                                                                                                                                  |
| Potting Material         |                         | Silicone (UL 94 V-0 rated)                                                                                                                                                |
| Pin Material             |                         | Copper Alloy (C2700)                                                                                                                                                      |
| Housing Type             |                         | Plastic Case                                                                                                                                                              |
| Mounting Type            |                         | PCB Mount                                                                                                                                                                 |
| Connection Type          |                         | THD (Through-Hole Device)                                                                                                                                                 |
| Footprint Type           |                         | SIP3                                                                                                                                                                      |
| Soldering Profile        |                         | Lead-Free Wave Soldering                                                                                                                                                  |
|                          |                         | 260°C / 5 s max.                                                                                                                                                          |
| Weight                   |                         | 2.6 g                                                                                                                                                                     |
| Thermal Impedance        | - Case to Ambient       | 41.0 K/W typ.                                                                                                                                                             |
| Environmental Compliance | - REACH Declaration     | <a href="http://www.tracopower.com/info/reach-declaration.pdf">www.tracopower.com/info/reach-declaration.pdf</a>                                                          |
|                          |                         | REACH SVHC list compliant                                                                                                                                                 |
|                          | - RoHS Declaration      | REACH Annex XVII compliant<br><a href="http://www.tracopower.com/info/rohs-declaration.pdf">www.tracopower.com/info/rohs-declaration.pdf</a><br>Exemptions: No Exemptions |

All specifications valid at nominal voltage, resistive full load and +25°C after warm-up time, unless otherwise stated.

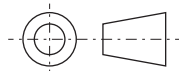
### Additional Information

|                            |                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------|
| Supporting Documents       | <a href="http://www.tracopower.com/overview/tsr3n">www.tracopower.com/overview/tsr3n</a>       |
| Frequently Asked Questions | <a href="http://www.tracopower.com/glossary-faq">www.tracopower.com/glossary-faq</a>           |
| Glossary                   | <a href="http://www.tracopower.com/info/glossary.pdf">www.tracopower.com/info/glossary.pdf</a> |
| Service Portal             | <a href="http://www.tracoserviceportal.com">www.tracoserviceportal.com</a>                     |

### Outline Dimensions



Dimensions in mm  
 General tolerances:  $\pm 0.5$   
 Pin tolerances:  $\pm 0.1$



### Pinout

| Pin | Function |
|-----|----------|
| 1   | +Vin     |
| 2   | Ground   |
| 3   | +Vout    |

Avoid routing PCB traces under the converter.